Filter Wheel Design for Optimal Color Reconstruction

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Abstract.
MAKE A NICE (WIDE) GRAPHIC TO PUT UNDER ABSTRACT
GET THE PHOTO OF ROSCO GELS OFF FRONT PAGE

Cameras' spectral sensitivity functions do not properly reconstruct the behavior of the human eye. This can lead to inaccurate color reproduction in images, especially under certain lighting conditions. One solution to this problem is to use a filter wheel with a set of filters that can be combined to reconstruct specific color matching functions. We design a framework for selecting filters from a given set to optimize for accurate color reconstruction, and apply this framework to the Rosco Supergel filter set. We also generealize our framework for other color matching functions beyond the XYZ functions.

1 Introduction

We aim to improve upon the classic filter wheel and monochromatic camera model. We simulate a system of captures which then reconstructs the XYZ color pallette with minimal error from a set of filter patches. Prior work has been limited by computational resources, and generally relies on heuristic measures to select filters. We provide a minimal error reconstruction for the set of Rosco Supergel[1] filters, and compute the relationship between number of filters N and minimum δE under standard illuminants.

Further, we provide a framework for filter selection for any target spectral response function. By generalizing our target function, we allow for applications beyond RGB reconstruction, such as hyperspectral imaging and multispectral imaging, or even non-imaging applications such as spectral sensing.

2 Method

2.1 Data Collection

We take the standard CIE 1931 XYZ color matching functions[2] as our target spectral response functions for reconstruction.

For our filter set, we use a standard set of Rosco Supergel[1] filters and measure their spectral response functions using a spectrophotometer under [FILL IN MODEL] lighting. We construct a dataset spanning λ 400-750 at 1nm intervals, and normalize the response functions to a maximum of 1.

Figure 1: Filter measurement setup with spectrophotometer and Rosco Supergel filters.

2.2 Filter Selection

Our search method is a brute-force combinatorial search over all possible filter combinations of size N from our filter set. For each combination, we compute the linear least squares solution (RMSE) to find the optimal weights for reconstructing the XYZ functions from the selected filters.

We minimize for the distance between the target spectral response functions and the reconstructed functions from the selected filters:

$$\arg\min_{\vec{i},\boldsymbol{M}} \sum_{X,Y,Z} F(p_C(\lambda) - \boldsymbol{M}_{\vec{i}} \int_{\lambda} I(\lambda) R(\lambda) e_{\vec{i}}(\lambda)\, d\lambda) \tag{1}$$

Table 1: Notation and symbols used in the text

Symbol	Description
N	number of filters
F	loss function
$I(\lambda)$	Illuminant intensity
$e_i(\lambda)$	spectral transmission function
$R(\lambda)$	surface reflectance spectrum
$q_c(\lambda)$	camera spectral sensitivity
$p_c(\lambda)$	target transmission function
c	color $c \in \{R, G, B\}$
$\boldsymbol{M}$	$3 \times N$ matrix of weights
$\vec{i}$	a row vector of filter indices
$\boldsymbol{I}$	integral in optimizer above

To solve this, we can express the problem in matrix form for each filter index subset i and each channel c:

$$\arg\min_{\boldsymbol{M_i}} ||\boldsymbol{P} - \boldsymbol{M_i I}||^2 \tag{2}$$

For F as Mean Squared Error (MSE), The optimal weights can be found using the pseudo-inverse:

$$\boldsymbol{M_i} = \boldsymbol{P I}^T (\boldsymbol{I I}^T)^{-1} \tag{3}$$

And for other F, such as SMAPE, we use gradient descent to find the optimal weights, informing our initial guess with the MSE solution.

We find the optimal filter combination for each N by selecting the combination with the lowest loss F. We plot the errors across all combinations for N=4:

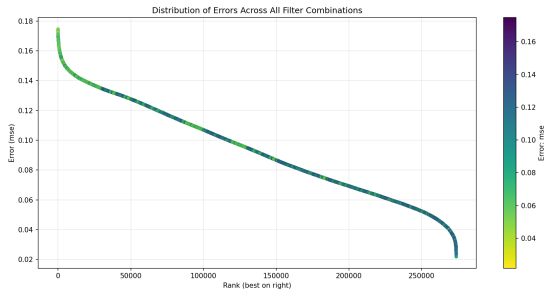

Figure 2: MSE Error distribution across all filter combinations for N=4

As the search space grows combinatorially with N, we consider N up to 10, which is a reasonable number of filters for a filter wheel system. In filter wheels consisting of 10 or more filters, we utilize a greedy heuristic to add candidate filters to our search space, selecting filters that minimize the error when added to the best combination from $N - 1$ filters.

3 Evaluation

We evaluate our method by comparing the reconstructed XYZ functions from our selected filters to the target XYZ functions, and computing the color difference ΔE under standard illuminants. We use the Macbeth Color Checker [3] as our test set, and evaluate under the D65 illuminant. For each number of

filters N, we compute the minimum ΔE across all combinations of N filters from the Rosco Supergel set, and plot the relationship between N and ΔE.

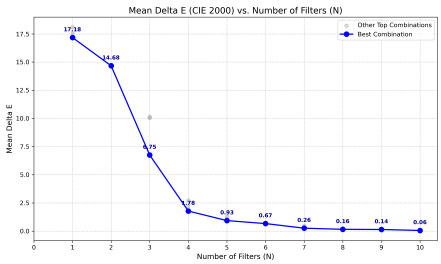

Figure 3: Best ΔE for each number of filters N

We also plot the resultant reconstructed XYZ functions for the best filter combination for various results at N filters, and compare it to the target XYZ functions. The $N = 3$ case, we can see visible artifacts in the transmission functions. Note that the best MSE solution does not necessarily correspond to the best ΔE solution, and this is most evident at $N = 3$ where the 3'rd best MSE achieves the low ΔE of 6.75.

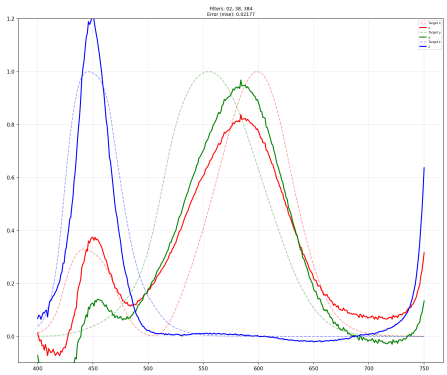

Figure 4: Spectral Transmission for best XYZ Reconstruction for $N = 3$

The best filter combination for $N = 3$ achieves a ΔE of 6.7, which is above the just noticeable difference threshold of 2.3, indicating that the reconstructed colors may be distinguishable from the target colors under certain conditions.

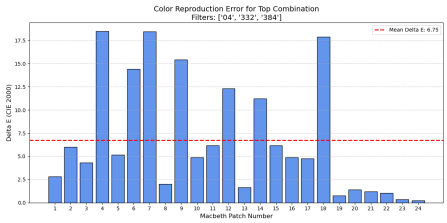

Figure 5: Best ΔE for each number of filters $N = 3$

However, $N = 4$ achieves a ΔE of 1.7, which is below the just noticeable difference threshold, indicating that the reconstructed colors are likely indistinguishable from the target colors under typical viewing conditions.

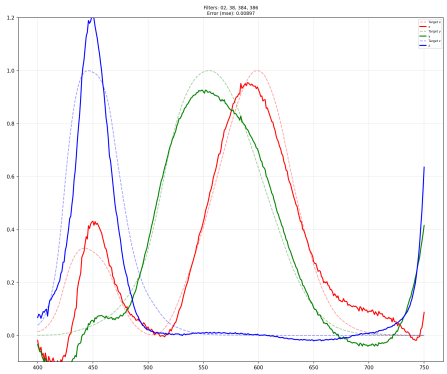

Figure 6: Spectral Transmission for best XYZ Reconstruction for $N = 4$

Further, $N = 10$ achieves a ΔE of 0.08, virtually identical to the target patches.

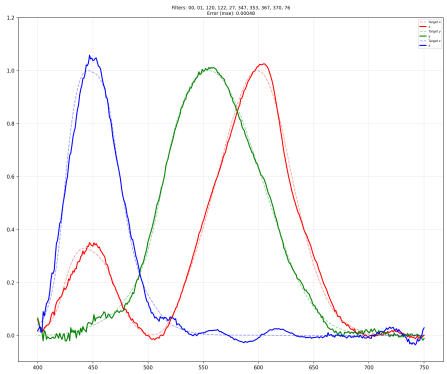

Figure 7: Spectral Transmission for best XYZ Reconstruction for $N = 10$

The distribution of ΔE across the 24 Macbeth Color Checker patches for the best filter combination for $N = 10$ shows that all patches achieve a ΔE below 0.1.

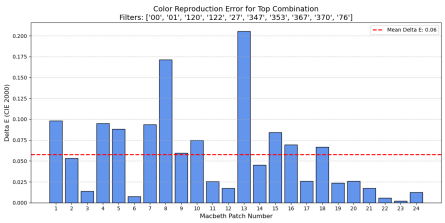

Figure 8: Best ΔE for each number of filters N

We show a render of the 24 Macbeth Color Checker patches under the D65 illuminant, reconstructed from the best filter combination for $N = 10$, and compare it to the target Macbeth Color Checker render (upper left/lower right corners respectively).

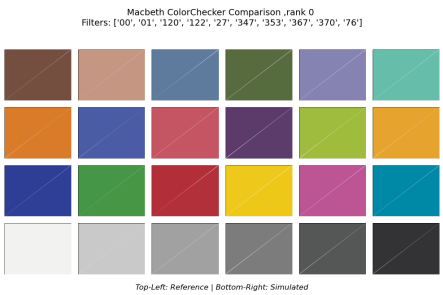

Figure 9: Macbeth Color Checker Comparison for $N = 10$

In comparison, the best filter combination for $N = 3$ shows visible color differences in the render of the Macbeth Color Checker patches, particularly in the red and blue patches.

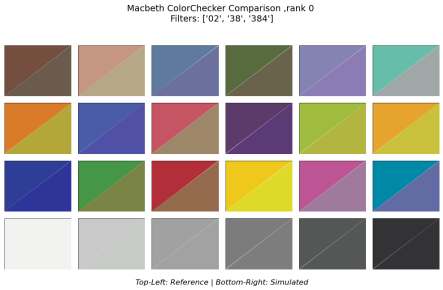

Figure 10: Macbeth Color Checker Comparison for $N = 10$

In addition, the 3×10 $\boldsymbol{M}$ matrix for the best $N = 10$ reconstruction is shown below, displaying two important characteristics. First, all 10 filters are 'active' in at least one dimension at all times.

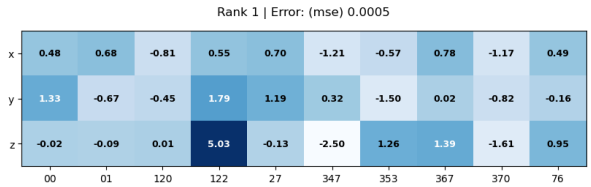

Figure 11: $\boldsymbol{M}$ matrix for $N = 10$ reconstruction

4 Conclusion

We are able to achieve fairly accurate reconstructions of the XYZ color matching functions with as few as 4 filters, and near perfect reconstructions with 10 filters. This suggests that a filter wheel system with 10 filters could be sufficient for high-quality color reconstruction. This system does have drawbacks, such as shutter speed on the order of seconds and the need for a static scene.

References

[1] Rosco Laboratories, "Rosco mycolor." `https://legacy.rosco.com/mycolor/mycolor.cfm`, 2026. Accessed: 2026-03-29.

[2] T. Smith and J. Guild, "The C.I.E. colorimetric standards and their use," *Transactions of the Optical Society*, vol. 33, pp. 73–134, Jan. 1931.

[3] D. Pascale, "Rgb coordinates of the macbeth colorchecker," 2006.